

CO1 AT3

Name: MANN

Reg No: 192311090

1. Find the Second Largest Element in an Array

```
START
READ n
READ array elements
largest = -∞
second = -∞
FOR each element x
    IF x > largest
        second = largest
        largest = x
    ELSE IF x > second AND x != largest
        second = x
    END IF
END FOR
PRINT second
STOP
```

2. Check Whether a Number is an Armstrong Number

```
START
READ n
temp = n
sum = 0
count = number of digits in n
WHILE temp > 0
    digit = temp MOD 10
    sum = sum + digit^count
    temp = temp DIV 10
END WHILE
IF sum == n
    PRINT "Armstrong Number"
ELSE
    PRINT "Not an Armstrong Number"
END IF
STOP
```

3. Find the Frequency of Each Element in an Array

```
START
READ n
READ array elements
FOR i = 0 TO n-1
    count = 0
    FOR j = 0 TO n-1
        IF array[i] == array[j]
            count = count + 1
        END IF
    END FOR
    IF element is not already processed
        PRINT array[i], count
    END IF
END FOR
STOP
```

4. Check Whether Two Strings are Anagrams

```
START
```

```

READ string1
READ string2
IF length(string1) != length(string2)
    PRINT "Not Anagram"
    STOP
END IF
SORT string1
SORT string2
IF string1 == string2
    PRINT "Anagram"
ELSE
    PRINT "Not Anagram"
END IF
STOP

```

5. Find the Sum of Both Diagonals of a Matrix

```

START
READ n
READ n × n matrix
sum1 = 0
sum2 = 0
FOR i = 0 TO n-1
    sum1 = sum1 + matrix[i][i]
    sum2 = sum2 + matrix[i][n-1-i]
END FOR
IF n is odd
    total = sum1 + sum2 - matrix[n/2][n/2]
ELSE
    total = sum1 + sum2
END IF
PRINT total
STOP

```